

Question 1

Variables and Parameters. Consider Newton's cooling model

$$\frac{dT}{dt} = -k(T - T_a).$$

Identify the independent variable, the dependent variable, the parameters and the state variable.

Question 2

Dimensional Consistency. Determine whether

$$s = ut + at$$

is dimensionally consistent.

Question 3

Population Growth Model. Suppose the population $P(t)$ grows at a rate proportional to its size. Construct the differential equation, and the equation.

Question 4

Water Tank Model. A tank initially contains 200 L of water. Water enters at 10 L/min and leaves at 4 L/min. Construct the governing differential equation and solve the equation.

Question 5.

A particle is moving on a straight line with constant acceleration a . If x is the distance from some fixed point on the straight line, show that the equation

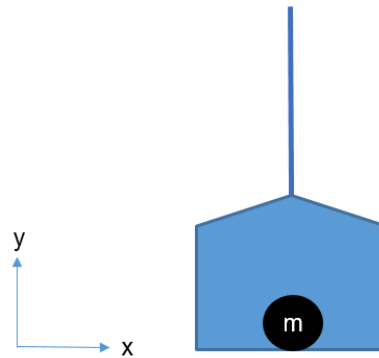
$$\frac{d^2x}{dt^2} = a$$

can be written as $v \frac{dv}{dx} = a$, where v is the particle's speed. If $v = v_0$ at $x = 0$, integrate this equation and show that $v^2 = v_0^2 + 2ax$.

Question 6

A man 70 kg, stands on the floor of a lift.

(a) The force acting on the mass includes the gravitational force and the reaction from the floor. Show the directions of all forces acting on the mass. In the coordinate shown, what is the total force acting on the mass?

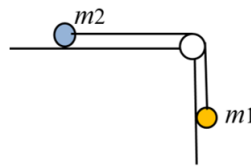


(b) Suppose the lift moves upwards with an acceleration of 4 m/s^2 . Use Newton's 2nd law to calculate the reaction of the floor (the force acting on the man from the floor).

(c) Suppose the lift moves downwards with acceleration of 4 m/s^2 . Find the reaction of the floor.

Question 7

A mass of m_2 kg is placed on a smooth horizontal table, and connected by a light inextensible string passing over a small smooth pulley at the edge to a mass of m_1 kg hanging freely.



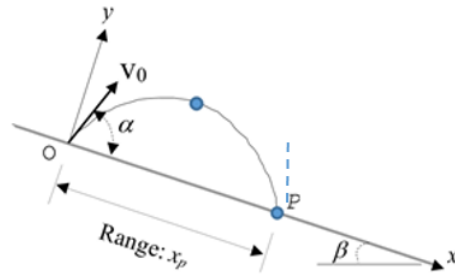
(a) Assume the tension in the string is T . Show the directions of all forces acting on m_1 . Then establish the equation of motion for m_1 by using Newton's 2nd law (take $\uparrow$ or $\downarrow$ for a positive direction for both forces and acceleration).

(b) Show the directions of all forces acting on m_2 . Then establish the equation of motion for m_2 by using Newton's 2nd law (take $\rightarrow$ or $\leftarrow$ as a positive direction for both forces and acceleration).

(c) The equations for m_1 and m_2 constitute a system of two equations with two unknowns tension force T and acceleration a . Solve the system to find T and a .

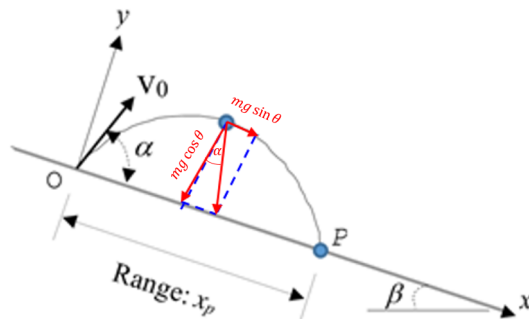
Question 8.

A catapult projects a stone in the normal direction to a playground which slopes at the angle of 10° to the horizontal. The initial speed of the stone is 18 m/s . Ignoring air resistance, calculate the range parallel to the playground.



Question 9.

From Question 8, suppose that the playground is inclined at an angle α to the horizontal. If the stone were to be fired with velocity v at an acute angle β to the playground, in the vertical plane through the line of greatest slope, find the maximum range parallel to the playground.



End of Questions